Physics-Informed Neural Networks for Mechanics: Reproduction, Diagnosis and Improvements

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github.com/DBS13062006/pinn-mechanics

1. Problem statement

A physics-informed neural network (PINN) learns a function that obeys a differential equation: automatic differentiation gives the network's derivatives, so the equation's residual can be added to the loss. Our base paper (Zhang, Chen and Yang, Stanford CS229, 2020) applies PINNs to three problems in mechanics: **(4.1)** consolidation of a soil layer, **(4.2)** steady heat conduction in a square plate, and **(4.3)** discovering the damping c and stiffness k of an oscillator from measurements. It reports *one* training run per problem and lists the coupled poromechanics equations as future work. We asked: **(i)** do the results hold across random seeds? **(ii)** where does the remaining error sit, and why? **(iii)** can it be removed? **(iv)** does the method extend to the paper's future-work problem?

2. Dataset

A PINN needs no external dataset: it learns from the governing equations. Its "data" are *collocation points* where the equation is enforced, boundary and initial points, and, for the inverse problems, *synthetic measurements* generated from exact solutions. Every set is drawn with a fixed seed, so all runs are repeatable, and every problem has an exact reference solution checked against an independent second method, so every error is measured exactly.

Problem	Domain	Training points	Measurements (synthetic)	Exact reference / check
4.1 Consolidation	$x \in [0,1]$, $t \in [0, 0.5]$	500 PDE, 250 BC, 125 IC	none	erfc images / Fourier
4.2 Heat	$(x,y) \in [-1,1]^2$	1500 PDE, 500 BC	none	series / finite diff.
4.3 Oscillator	$t \in [0,15]$	100 ODE, IC	50 equispaced $u(t)$ with $c=0.4$, $k=4$; noise study: 0–10 % noise, 15 points, $t \le 5$ only	closed form / ODE solver
5 Poromechanics	$x \in [0,1]$, $t \in [0,1]$	600 PDE, 150 per BC, 150 IC	20 surface-settlement readings, 1 % noise, $c=0.5$	derived erfc / series

Accuracy is measured on dense test sets (a 201×201 grid, or 10 000 points for the oscillator) and by the PDE residual on *fresh* points never used in training.

3. Approach

Reproduce. We rebuilt each experiment with the paper's exact settings (network size, tanh, Glorot initialisation, point counts, loss weights, learning rate, Adam then L-BFGS) and ran it over 5–10 seeds. The loss is $\mathcal{L} = w_f\overline{r^2} + w_b\overline{(Bu)^2} + w_d\overline{(u-u^*)^2}$; in inverse problems the unknown constants are trained with the network weights. **Diagnose.** We located the error in space and time and traced each to its cause. **Fix.** We built the known local behaviour into the network's *output* rather than the loss:

- **Oscillator: two-stage warm start.** Fit the data alone, read c, k off the fitted curve by least squares on its derivatives, train with them frozen, then release them. Cause fixed: the network learns the curve left to right, and a still-flat tail looks like heavy damping.
- **Consolidation: corner enrichment.** $p = \mathrm{erf}\left(x/2\sqrt{t}\right) + x\,t\,N(x,t)$. The erf term carries the infinitely steep front at $x = 0, t = 0$ that no smooth network can represent.
- **Heat: exact edges + corner terms.** $T = (1-x^2)(1-y^2)N + \sum_{\text{corners}} u_c$, where u_c is the exact local solution of a 90° corner, derived by hand (T_{xx} is direction-dependent there).
- **Poromechanics (paper §5).** One PINN with two outputs, displacement U and pore pressure P, trained on the *coupled* Biot equations (5.1)–(5.2) in 1D, with the corner enrichment carried over to both fields. Forward: solve for U, P. Inverse: discover the consolidation coefficient c from settlement readings.

We also benchmarked parameter discovery on noisy and sparse data against a classical ODE fit and the Cramér–Rao bound, which the paper did not test.

4. Implementation

Python 3.12 and PyTorch, in float64 (L-BFGS needs it near convergence). The code has three layers: `pinn/` is a problem-agnostic engine (tanh MLP with input scaling, autograd derivatives, an Adam→L-BFGS trainer that logs every loss term and rolls back if L-BFGS diverges, adaptive sampling); `problems/` holds one module per physics problem (residuals, boundary conditions, exact solutions, output transforms); `experiments/` has one script per study that writes its figure and tables. 57 unit tests check the engine, every exact solution and the corner facts

without training, and run on Linux and Apple Silicon in GitHub Actions. `demo.py` replays every comparison live in about a minute per problem on a laptop CPU.

5. Results and conclusions

Problem	Paper's method, reproduced (median over seeds)	Our fix	Gain
4.3 Oscillator (c, k)	3/5 seeds within 1 %, 699 s per run	two-stage: **10/10 seeds**, 23 s	reliable, 30× faster
4.1 Consolidation	rel. L2 error 1.4e-2, pressure overshoots to 1.11	enriched: **6.4e-5**, no overshoot	≈200× lower
4.2 Heat	max error 3.8e-3 (on the edges)	exact BC + corners: **2.7e-5**	≈140× lower
5 Poromechanics, forward	rel. L2 error of P: 2.1e-1	enriched: **1.3e-3**	coupled U, P solved
5 Poromechanics, inverse	0/5 seeds find c within 1 %	**5/5**, median error 1e-3 with 1 % noise	c discovered

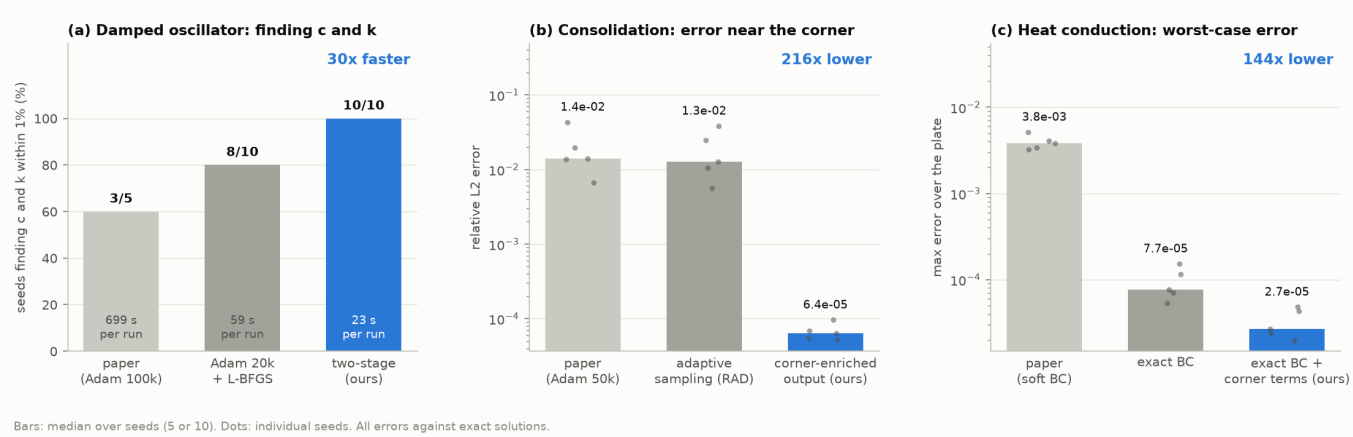


Figure 1: Paper's method (grey) vs our fixes (blue). Bars: median over seeds; dots: individual seeds.

1. **One run is not evidence.** The paper's accuracy holds for some seeds and not others.
2. **Training loss can mislead.** Near corners the loss looked converged while the equation was badly violated; residuals on fresh points exposed it every time.
3. **Put known structure in the output, not the loss.** Exact boundary conditions and analytical enrichment beat adaptive sampling and longer training, and the same idea solved the paper's future-work problem.
4. **Warm starts make inverse problems reliable**, from 3/5 to 10/10 seeds at 30× the speed.
5. **A PINN is not always the right tool.** With a known closed form and two parameters, a classical fit matches or beats it; PINNs earn their place when data are incomplete and physics must fill the gaps.

Limitations. Enrichment terms need analytical insight into each problem; uncertainty estimates for discovered parameters are future work. Full details are in the 16-page report and the repository README.